Holy Cross College

Semester 1 Examination 2015

Question/Answer Booklet

Year 10 Science (General)

Please place your student identification label in this box

Student Name SOLUTIONS

Student's Teacher

Time allowed for this paper

Reading time before commencing work: 10 minutes
Working time for paper: 100 minutes

Materials required/recommended for this paper

To be provided by the supervisor

This Question/Answer Booklet
Data Sheet

To be provided by the candidate

Standard items: pens, pencils, eraser, correction fluid, ruler, highlighters.

Special items: non-programmable calculators satisfying the conditions set by the School Curriculum and Standards Authority for this course.

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised notes or other items of a non-personal nature in the examination room. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Suggested working time (minutes)	Marks available	Percentage of exam
Section One: Multiple Choice	55	55	35	55	37
Section Two: Short Answer	13	13	45	~~82~~ 81	56
Section Three: Comprehension	1	6	20	10	7
				~~147~~ 146	100

Instructions to candidates

1. The rules for the conduct of examinations at Holy Cross College are detailed in the College Examination Policy. Sitting this examination implies that you agree to abide by these rules.

2. Write your answers in this Question/Answer Booklet.

3. Working or reasoning should be clearly shown when calculating or estimating answers.

4. You must be careful to confine your responses to the specific questions asked and to follow any instructions that are specific to a particular question.

5. Spare pages are included at the end of this booklet. They can be used for planning your responses and/or as additional space if required to continue an answer.

    - Planning: If you use the spare pages for planning, indicate this clearly at the top of the page.
    - Continuing an answer: If you need to use the space to continue an answer, indicate in the original answer space where the answer is continued, i.e. give the page number.
    - Fill in the number of the question(s) that you are continuing to answer at the top of the page.

SECTION 1 MULTIPLE CHOICE (55 marks)

Choose the best answer for each question in the Multiple Choice Question booklet.

Mark your selection with a cross.

1	~~(a)~~	(b)	(c)	(d)	31	(a)	~~(b)~~	(c)	(d)
2	~~(a)~~	(b)	(c)	(d)	32	(a)	(b)	~~(c)~~	(d)
3	(a)	~~(b)~~	(c)	(d)	33	(a)	(b)	(c)	~~(d)~~
4	(a)	(b)	(c)	~~(d)~~	34	(a)	(b)	(c)	~~(d)~~
5	(a)	(b)	(c)	~~(d)~~	35	(a)	~~(b)~~	(c)	(d)
6	(a)	(b)	~~(c)~~	(d)	36	~~(a)~~	(b)	(c)	(d)
7	(a)	~~(b)~~	(c)	(d)	37	~~(a)~~	(b)	(c)	(d)
8	(a)	~~(b)~~	(c)	(d)	38	~~(a)~~	(b)	(c)	(d)
9	~~(a)~~	(b)	(c)	(d)	39	(a)	(b)	(c)	~~(d)~~
10	(a)	(b)	(c)	~~(d)~~	40	~~(a)~~	(b)	(c)	(d)
11	(a)	~~(b)~~	(c)	(d)	41	~~(a)~~	(b)	(c)	(d)
12	~~(a)~~	(b)	(c)	(d)	42	~~(a)~~	(b)	(c)	(d)
13	~~(a)~~	(b)	(c)	(d)	43	(a)	(b)	(c)	~~(d)~~
14	(a)	~~(b)~~	(c)	(d)	44	(a)	(b)	~~(c)~~	(d)
15	~~(a)~~	(b)	(c)	(d)	45	(a)	(b)	~~(c)~~	(d)
16	(a)	(b)	~~(c)~~	(d)	46	(a)	(b)	(c)	~~(d)~~
17	(a)	~~(b)~~	(c)	(d)	47	(a)	(b)	(c)	~~(d)~~
18	(a)	(b)	~~(c)~~	(d)	48	(a)	(b)	(c)	~~(d)~~
19	(a)	~~(b)~~	(c)	(d)	49	(a)	(b)	(c)	~~(d)~~
20	(a)	(b)	~~(c)~~	(d)	50	~~(a)~~	(b)	(c)	(d)
21	(a)	(b)	(c)	~~(d)~~	51	~~(a)~~	(b)	(c)	(d)
22	(a)	(b)	(c)	~~(d)~~	52	(a)	~~(b)~~	(c)	(d)
23	(a)	~~(b)~~	(c)	(d)	53	(a)	~~(b)~~	(c)	(d)
24	(a)	~~(b)~~	(c)	(d)	54	~~(a)~~	(b)	(c)	(d)
25	(a)	~~(b)~~	(c)	(d)	55	~~(a)~~	(b)	(c)	(d)
26	~~(a)~~	(b)	(c)	(d)					
27	(a)	(b)	(c)	~~(d)~~					
28	~~(a)~~	(b)	(c)	(d)					
29	(a)	(b)	~~(c)~~	(d)					
30	(a)	~~(b)~~	(c)	(d)					

SECTION 2 SHORT ANSWER (~~82~~ 81 marks)

Attempt all questions. Write your answers in the spaces provided.

1. Write the correct number next to each definition.

1. Homozygous
2. Heterozygous
3. Allele
4. Chromosome
5. Gene
6. Recessive
7. Dominant
8. Phenotype
9. Genotype
10. Nucleus

DEFINITION	NUMBER
Part of the cell containing DNA.	10
Allele that is masked by another.	6
Unit of heredity.	5
Alternatives for a gene.	3
When there are 2 of the same allele for a particular gene.	1
There are 46 of these in a human cell.	4
The genes representing a characteristic.	9
The physical characteristic shown.	8
The allele that will mask another.	7
When there are 2 different alleles for a particular gene.	2

[$\frac{1}{2}$ mark each] (5 marks)

2. Carefully examine the photo of a young animal (centre) showing the genetic condition called albinism.

(a) Describe the ***phenotype*** of an animal with albinism.

(1 mark)

white fur

(b) The young lion cub inherits all its features from its parents. As you can see in the picture, the lion cub shows the condition but neither of its parents (the male and female lions in the picture) is albino.

Suggest ***two*** reasons that explain why the phenotype for albinism seen in the lion cub is not visible in either of the lion's parents.

(2 marks)

(i) Albino fur is a recessive characteristic.

(ii) Mutation occurred in one of the parents

• Gene for normal fur dominates (masks) the albino gene.

[Any suitable two reasons OK].

3. In guinea pigs, rough coat is dominant to smooth coat. Two hybrid rough coated guinea pigs are mated. Determine the genotype and phenotype of the possible offspring. Include percentages to show the probabilities. Show all of your working.

(4 marks)

Rr x Rr (1)
♂ ♀

♀ \ ♂	R	r
R	RR	Rr
r	Rr	rr

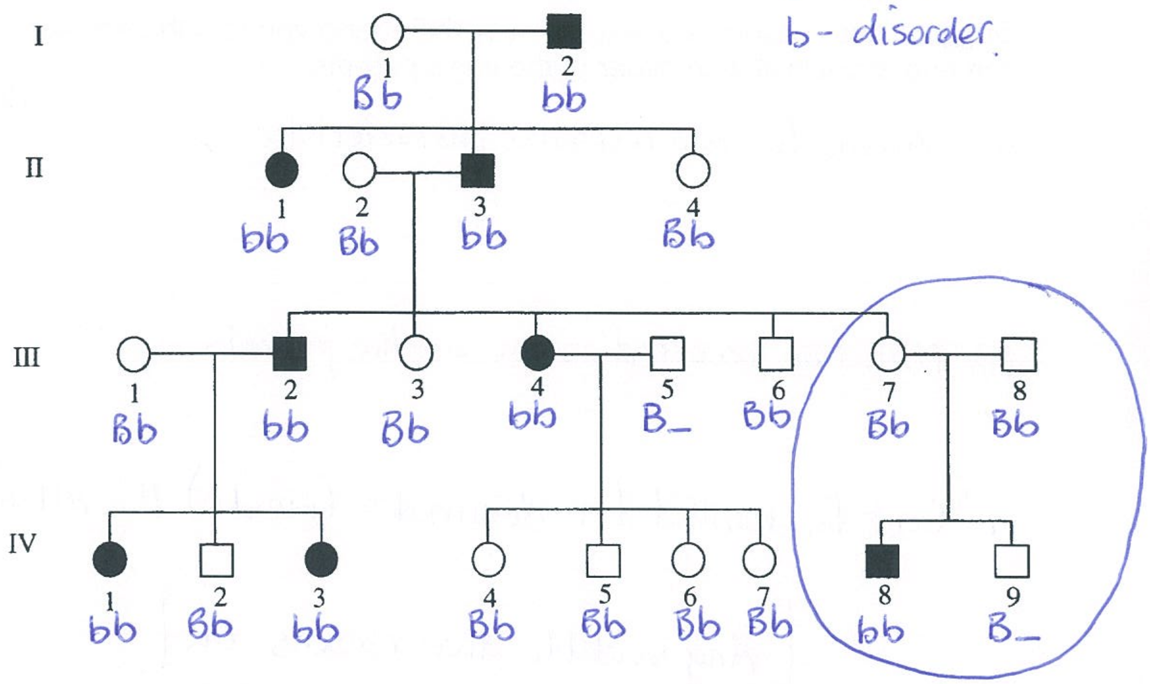
(1)

GENOTYPE $\underbrace{\text{RR Rr Rr}}$ $\underbrace{\text{rr}}$ (1)

PHENOTYPE Rough 75% Smooth 25% (1)

4. The chart below shows the inheritance of a rare (but not serious) blood disorder through four generations of a family. Individuals with the condition are represented by the shapes that have been shaded in.

B - normal
b - disorder



(a) How many people in total show the disorder in this family? 8
(1 mark)

(b) How many of the people with the disorder are female? 4
(1 mark)

(c) How many children did person II 2 and II 3 have in total? 5
(1 mark)

(d) Do you think the gene that causes the disorder in this family is controlled by a dominant gene or a recessive gene? Give one reason from the chart to explain your answer.
(***You may need to circle the part used for your explanation***.)
(2 marks)

Recessive. (1)

Parents don't have the disorder and one child does. (1)

(e) Assume the dominant gene for the disorder is **D**, and the recessive gene is **d**.

Give the genotype for the following individuals.

II 2 Bb   III 3 Bb   III 7 Bb   IV 8 bb
(4 marks)

[1 mark each]

5. In snapdragon flowers, red colour is ***incompletely dominant*** to white colour. Heterozygous plants produce pink flowers.

(a) Describe the genotype and phenotype of possible flower colours if a pink snapdragon is crossed with a white snapdragon. Include the ratios and show all of your working.

(4 marks)

RW x WW (1)

♂ ♀

	R	W
W	RW	WW
W	RW	WW

(1)

GENOTYPE $\underbrace{\text{RW RW}}$ $\underbrace{\text{WW WW}}$ (1)

PHENOTYPE 50% pink 50% white (1)

(b) In some plants, flowers show ***co-dominance***. For example, crossing a red flower with a white flower produces red flowers with white parts, or vice versa. Explain how co-dominance is different to incomplete dominance.

(2 marks)

Co-dominance - both characteristics show
e.g. red + white show (1)

Incomplete dominance - a new characteristic is seen. (1)
e.g. pink colour

6. Mitosis and meiosis are both types of cell division that occur in humans and other organisms.

Complete the following table comparing mitosis and meiosis ***in humans***.

	Mitosis	Meiosis
Number of cell divisions	1	2
~~Number of DNA replications~~		
Number of daughter cells produced	2	4
Number of chromosomes in parent cell	46	46
Number of chromosomes in daughter cells	46	23
For growth and development? (Yes / No)	Yes	No
For reproduction? (Yes / No)	No	Yes

[½ mark each] (~~7~~ 6 marks)

7. Answer 'true' or 'false' to each of the following statements about mitosis and meiosis.

	True / False
Mitosis occurs in the nucleus of a cell.	T
Mitosis results in the formation of one new cell.	F
Two new and identical cells form as the result of mitosis.	T
In mitosis New cells contain the same number of chromosomes as the original cell.	T
Meiosis is a process that results in the production of gametes.	T
Meiosis occurs in two stages, producing two cell divisions.	T
New cells produced by meiosis have the same chromosomal number as the original cell.	F

[1 mark each] (7 marks)

8. (a) Who determines the sex of any children born into a family - the father or mother? Give a reason for your answer.

(2 marks)

Father (1)

Father is XY and the Y chromosome makes an individual a male. (1)

(b) A family has three girls born over five years and the mother is pregnant with a fourth child. What is the chance of the unborn child being a female?

Justify your answer by setting out a genetics calculation using a Punnett square.

(4 marks)

XY × XX (1)
♂ ♀

♀ \ ♂	X	Y
X	XX	XY
X	XX	XY

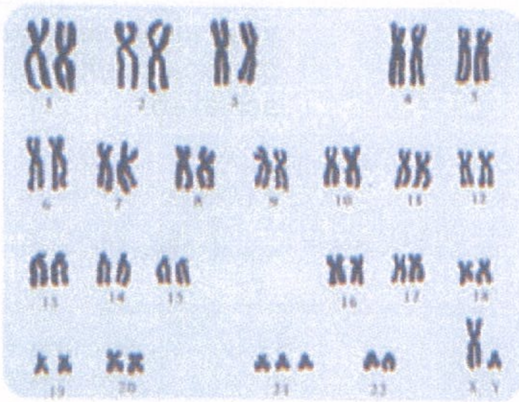
(1)

GENOTYPE $\underbrace{\text{XX XX}}$ $\underbrace{\text{XY XY}}$ (1)

PHENOTYPE 50% female 50% male

∴ Chance is 50% female (1)

9. A newborn baby shows distinct facial abnormalities. A karyotype was prepared to determine whether there were any chromosomal abnormalities. The karyotype is shown to the right.



(a) What is the total number of chromosomes shown?

47 (1 mark)

(b) Is the child male or female? How do you know?

(2 marks)

Male. (1)

Has XY chromosome. (1)

(c) As the geneticist, what would you advise the parents about the health of their baby?

(2 marks)

Child will have some abnormality - possibly (1)
Down's Syndrome.

There is an extra chromosome at position 21. (1)

10. Astronomers have found an Earth-like planet orbiting a star called Mu Arae. It takes light 50 years to reach Earth from this star, travelling at 300,000 km/second.

(a) The best speed possible for a spaceship with our current technology is about 50,000 km/h, which is equal to about 15 km/second.

Do you think humankind will be able to get to Mu Arae in a reasonable time? ***Give a detailed reason for your answer.***

(2 marks)

- No - it would take far too long. (1)
- The time will be thousands of years and we would die before getting there. (1)

(b) Describe ***ONE*** possible way in which humankind might be able to overcome the vast distances required to travel to nearby galaxies. It may be a theoretical or practical method, but you must give some details about how it might be achieved.

(3 marks)

- Freeze people and revive them when we get there. Suspended animation. Computers run everything.
- Use a worm-hole to shorten the distance. Fold space on itself.

[Can be any reasonable attempt with some explanation.]

11. Stars begin life with the same beginning - a protostar of hydrogen gas forms under gravity and generates heat to start a nuclear reaction. Once formed, it is the size of the star that determines its ultimate pathway to its "death".

(a) Our Sun is an average size, and will continue to produce energy for how many more years?

5 billion years (1)

(1 mark)

(b) Describe the stages involved in the lifecycle of a star similar in size to our Sun. Consider the stages once the star has ***started to produce sustained energy***. ***You must name each stage and give a brief description of what is occurring.***
(HINT: There are 4 stages.) (4 marks)

Sun
↓
Red giant - star expands as fuel runs out
↓
Nova - outer layers explode off as the star collapses.
↓
White dwarf - burns hot and brightly; no nuclear reactions.
↓
Blackdwarf - cold and dark, all heat energy lost.

[Name = ½ mark Description = ½ mark].

(c) Very large stars undergo a ***supernova*** towards the end of their life.

(i) What is a supernova?

(1 mark)

- Large star explodes off its outer layers.

(ii) Describe what happens to the star to cause the supernova to occur.

(2 marks)

- Star swells to a supergiant as fuel runs out. (1)
- Core collapses and outer layers explode off. (1)

(d) After a supernova, these large stars become black holes. Describe ***two characteristics*** of a black hole.

(2 marks)

- Incredibly dense. (1)
- Huge gravitational field - light can't escape. (1)

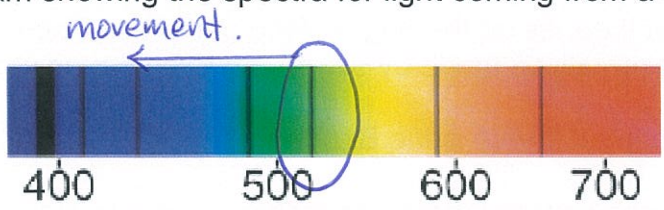
(e) Describe ***one way*** in which scientists might detect the presence of a black hole in a region of space.

(1 mark)

- Effect on nearby stars or planets. (1)
- Distortion of light from distant galaxies or stars.

[Any reasonable point.]

12. Below is a diagram showing the spectra for light coming from a star.



(a) What causes the dark lines in the spectra?

(2 marks)

- Light from the star is absorbed by its atmosphere. (1)
- Each element absorbs certain frequencies. (1).

(b) What is a "blue shift"? Use the diagram to help explain your answer.

(2 marks)

- Spectral lines appear to move towards the blue end. (1)

[Use of diagram - 1 mark].

(c) What does your answer to (b) tell astronomers about the movement of a star or galaxy relative to Earth?

(1 mark)

- Movement is towards Earth.

(d) Describe what Edwin Hubble discovered about distant galaxies when he observed the light coming from them.

(2 marks)

- Red-shifted. (1)
- Furthest galaxies show greatest red-shift. (1).
  ⇒ moving fastest.

13. There are several theories on the origin of the universe. Each has its merits in explaining the observations made by astronomers on the universe as it appears now.

The currently accepted theory is the Big Bang Theory.

(a) Briefly describe how this theory describes the origin of the universe. Include a description of what was present at the beginning.

(3 marks)

- Singularity at beginning. (1)
- Consisted of dense energy only. (1)
- Expands rapidly into space and time. (1)
- Can include matter being created from energy. (1)

[Any 3 pertinent points.]

(b) How does Einstein's equation ($E = mc^2$) relate to this theory?

(2 marks)

- It relates energy and matter. (1)
- After the initial expansion, energy changes into matter. (1)

(c) Describe ***two pieces*** of evidence that supports the theory.

(2 marks)

- Distant galaxies are red-shifted - moving away at fastest speeds. (1)
- Microwave radiation - shows the remnants of the expansion. Galaxies not evenly spread. (1)
- Expansion appears to come from one point. (1)

[Any 2 reasonable points.]

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SECTION 3 COMPREHENSION (10 marks)

Read the article and then answer the questions that follow. Write your answers in the spaces provided.

Mutation

A mutation is a change in a gene or chromosome in a cell that can be passed on to daughter cells produced by mitosis or meiosis of this cell. Mutation is the process by which a gene (or chromosome) changes structurally and therefore the phenotype changes as a result. Without mutation and the alleles it produces, the biological diversity that exists today would not have evolved.

Mutations, put simply, are the permanent alterations in a cell's DNA that affect the nucleotide sequence of one or more genes. They can be helpful or harmful to a species, but are usually harmful. The simplest is a point mutation, the change in a single nucleotide. Mutations are most likely to involve cells undergoing cell division. A single cell, a group of cells or an entire individual may be affected. An entire individual may be affected if mutations occur during meiosis or early in development. For example, a mutation that affects a gamete will be inherited by a child of that individual.

The rate of gene mutation is low. However, as each individual has a large number of genes, mutations are constantly occurring within a species. The rate at which mutations occur can be increased through exposure to mutagens such as radiation, chemicals or temperatures above the normal range for the organism.

Variation and evolution

Mutation ultimately provides genetic diversity; this is the raw material of evolution. Mutation rates are usually very low and, even if different alleles mutate at different rates, this by itself will not lead to rapid changes in populations.

Why would a diploid organism take a random sample of its DNA and combine it with a random
sample of someone else's DNA to produce offspring? The asexual way of life seems like an efficient way to reproduce yourself. What is the evolutionary benefit to organisms to develop sexual processes? Why is it better to combine the gametes of two individuals to produce a new generation of offspring?

Sexual reproduction allows for the provision of much more variation in organisms. Not only is just
half the genome of an individual passed on to every organism produced sexually, but that half is a random jumble, which may provide some advantage to the offspring. A sexual organism can achieve a significant amount of variation through recombination of DNA and fertilisation. In a changing environment a sexually reproducing organism increases the likelihood that it will adapt to changes that occur.

A haploid asexual organism accrues mutations as they occur over time in an individual. A sexually reproducing organism can combine several beneficial mutations in each generation by recombination and fertilisation. Therefore, sexually reproducing organisms can adapt at a much more rapid rate than asexually reproducing organisms.

Sexually reproducing organisms can eliminate harmful mutations each generation by either forming recombined offspring that are relatively free of mutation, or by producing offspring with many detrimental mutations that then die, removing the harmful mutations from the population.

1. What does the term "mutation" mean in genetics?

(2 marks)

- Change in a gene or chromosome. (1)
- Gene or chromosome changes structurally and phenotype changes. (1)
- Permanent alteration in a cell's DNA. (1)

[Any 2 OK].

2. What process has relied upon mutations to produce diversity and variation within a population?

(1 mark)

- Evolution. (1)

3. When might a mutation affect an entire individual?

(2 marks)

- If mutations occur during meiosis (1) or early in development. (1)

4. How is the rate of mutations within the population increased?

(2 marks)

- Exposure to mutagens. (1)

e.g. Radiation, chemicals, temperatures above normal. (1)

5. What provides genetic diversity within a population?

(1 mark)

• Mutation. (1)

6. Give ***two*** benefits of sexual reproduction within a population.

(2 marks)

• Achieves a significant amount of variation, allowing an organism to adapt to a changing environment. (1)

• Allows the organism to eliminate harmful mutations each generation. (1)

END OF EXAMINATION

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